

PROJECT 2: EXPLORATORY DATA ANALYSIS

DecodeLabs Industrial Training Kit

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Dataset: Dataset for Data Analytics (14 columns)

Date Range: January 2023 – June 2025

1. Problem Statement

The business has been operating for over two years (January 2023 – June 2025). Stakeholders need to understand:

1. What is the current state of the business?
2. Is revenue growing or declining?
3. What factors drive order value?
4. Where are the opportunities for improvement?

This EDA examines 1,200 orders across 14 attributes to uncover patterns, trends, and outliers that inform business decisions.

2. Methodology

Data Source

1,200 e-commerce orders (cleaned in Project 1). 14 columns including Order ID, Date, Customer ID, Product, Quantity, Unit Price, Shipping Address, Payment Method, Order Status, Tracking Number, Items in Cart, Coupon Code, Referral Source, and Total Price.

Tools & Techniques

Analysis	Method
Descriptive statistics	Mean, median, count, standard deviation, quartiles
Distribution	Five-number summary
Outlier detection	IQR method
Relationships	Pearson correlation coefficient (r)
Trends	PivotTables with date grouping (monthly, quarterly, yearly)
Patterns	Weekday distribution analysis

3. Key Findings

3.1 Descriptive Statistics

Metric	Quantity	Unit Price	Items in Cart	Total Price
Mean	2.95	\$356.41	5.49	\$1,053.97
Median	3.00	\$364.21	5.00	\$823.62
Min	1.00	\$11.39	1.00	\$11.39
Max	5.00	\$699.93	10.00	\$3,456.40
Q1	2.00	\$186.06	4.00	\$410.52
Q3	4.00	\$521.57	7.00	\$1,578.48
Std Dev	1.41	\$197.18	2.28	\$819.86

- **Finding 1:** Total Price is right-skewed.
 - Mean (\$1,053.97) > Median (\$823.62). The average is inflated by high-value orders. The median is the more accurate measure of typical order value.
- **Finding 2:** Quantity is symmetric.
 - Mean (2.95) $\approx$ Median (3.00). Customers typically order 3 units.
- **Finding 3:** Cart size exceeds order quantity.
 - Average items in cart: 5.49 vs. Average quantity ordered: 2.95.

3.2 Outlier Analysis

Using the IQR method: IQR for Total Price = \$1,167.96; Upper bound = \$3,330.42. 8 orders exceeded this threshold (0.67% of records).

Attribute	Value
Quantity	All = 5 (maximum)
Products	Printer, Laptop, Tablet, Chair, Monitor
Math check	All pass (Total = Quantity $\times$ Unit Price)
Classification	Legitimate high-value transactions
Average outlier value	\$3,379.05 (4.1 $\times$ the median)
Repeat customers	1 of 8

- **Finding 4:** The top 0.67% of orders average 4 $\times$ the median order value. These represent a VIP customer segment worth cultivating.
- **Finding 5:** 50% of outliers (4) were returned or cancelled. High-value orders have elevated return risk.

3.3 Correlation Analysis

Pair	r	R ²	Interpretation
Unit Price ↔ Total Price	0.72	0.52	Strong positive (pricing drives revenue)
Items in Cart ↔ Quantity	0.65	0.42	Strong positive (cart predicts purchase)
Quantity ↔ Total Price	0.62	0.38	Strong positive (bulk buying matters)
Items in Cart ↔ Total Price	0.39	0.15	Moderate positive
Quantity ↔ Unit Price	0.01	0.00	No relationship
Unit Price ↔ Items in Cart	0.00	0.00	No relationship

- **Finding 6:** Unit Price is the strongest driver of order value ($r = 0.72$). 52% of Total Price variance is explained by Unit Price alone.
- **Finding 7:** Quantity adds meaningful lift ($r = 0.62$). Each additional unit adds approximately \$358 to order value.
- **Finding 8:** Cart size predicts purchase behavior ($r = 0.65$). Customers who add more to cart tend to buy more.
- **Finding 9:** Price has no effect on quantity purchased ($r = 0.01$). Customers don't behave differently by price tier.

3.4 Time-Series Trends

Yearly Performance

Year	Orders	Revenue	AOV	Change
2023	510	\$552,643.24	\$1,083.61	-
2024	459	\$480,235.87	\$1,046.27	-13.1%
2025 (partial)	231	\$231,882.85	\$1,003.82	On pace for further decline

- **Finding 10:** Revenue declined 13.1% in 2024 vs 2023. Order count dropped (510 → 459) and average order value fell (\$1,083.61 → \$1,046.27).

Quarterly Performance

Quarter	2023	2024	Change
Q1	\$145,412.78	\$111,468.55	-23.3%
Q2	\$141,088.74	\$145,590.79	+3.2%
Q3	\$126,699.47	\$114,750.03	-9.4%

Q4	\$139,442.25	\$108,426.50	-22.2%
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- **Finding 11:** Three of four quarters declined YoY in 2024. The decline is systemic, not isolated to one quarter.
- **Finding 12:** Q4 2024 dropped 22.2% vs Q4 2023. The holiday season underperformed significantly.
- **Finding 13:** 2025 shows no recovery. Q1 2025 (-7.0%) and Q2 2025 (-11.9%) are both down vs 2024.

3.5 Weekday Distribution

Weekday	Orders	Revenue	AOV
Sunday	186	\$199,194.84	\$1,070.94
Monday	174	\$184,009.38	\$1,057.53
Tuesday	165	\$180,780.14	\$1,095.64
Wednesday	163	\$160,519.31	\$984.78
Thursday	166	\$182,066.56	\$1,096.79
Friday	174	\$190,598.20	\$1,095.39
Saturday	172	\$167,593.53	\$974.38

- **Finding 14:** Sunday has the highest order volume (186 orders, 15.5% of total).
- **Finding 15:** Thursday has the highest average order value (\$1,096.79).
- **Finding 16:** Weekday shoppers spend at least 4% more per order than weekend shoppers.
- **Finding 17:** Orders are evenly spread across the week. The difference between highest (186) and lowest (163) is only 14%.

4. Recommendations

4.1 Address the Revenue Decline

Problem: Revenue declined 13.1% in 2024 and is on pace for further decline in 2025.

Actions:

- Investigate customer acquisition
- Investigate retention (are repeat customers returning?)

4.2 Rebuild the Q4 Holiday Strategy

Problem: Q4 2024 dropped 22.2% vs Q4 2023.

Actions:

- Compare Q4 2023 vs Q4 2024 marketing spend
- Launch Q4 2025 campaign earlier (October instead of November)
- Build a holiday-specific product bundle

4.3 Improve Average Order Value

Problem: AOV has declined from \$1,083.61 (2023) to \$1,003.82 (2025).

Actions:

- Create product bundles to increase per-order value
- Upsell premium items at checkout (Unit Price has $r = 0.72$ with order value)
- Add free shipping thresholds to encourage larger carts
- Reduce reliance on coupons

4.4 Reduce Cart Abandonment

Problem: Average cart = 5.49 items, but average order = 2.95 units

Actions:

- Investigate checkout friction (steps, required fields, loading time)
- Add "You left items in your cart" email reminders
- Offer free shipping above a threshold

4.5 Target High-Value Weekday Shoppers

Problem: Weekday shoppers spend 4% more per order than weekend shoppers.

Actions:

- Schedule premium product campaigns for Tuesday–Friday
- Use weekends for volume plays (bundles, discounts)
- Since there is no strong seasonality, use always-on marketing

5. Summary Dashboard

Metric	Value
Total Revenue	\$1,264,761.96
Total Orders	1,200
Overall AOV	\$1,053.97
Date Range	Jan 2023 – Jun 2025
2024 Revenue Decline	–13.1%
Q4 2024 Decline	–22.2%
Best Month	Jun 2024 (\$68,068.54)
Best Weekday (Volume)	Sunday (186 orders)
Highest AOV Day	Thursday (\$1,096.79)
Strongest Correlation	Unit Price ↔ Total Price ($r = 0.72$)
Outlier Orders	8 (all signal)

6. Conclusion

The business is contracting, not growing. Revenue declined 13.1% in 2024, and 2025 shows no recovery. Average order value is eroding across every quarter and year.

However,

- Pricing power is real ($r = 0.72$)
- Cart abandonment is fixable
- June is a repeatable success (mid-year demand spikes consistently)